



Inspiring Excellence

Assignment : 1

Course Code : CSE340

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Ans to ques no-01

In Computer architecture the term "^{Dependability}~~Dependency~~ via Redundancy" means to keep the computer as fast possible while making sure the process is reliable. It is like having backup of the component that is keeping the computer fast. It increased the reliability by reducing any malfunction.

Ans to ques - 02

① Radius of silicon ignot is similar to each wafer. So, wafer Area = $\pi(R)^2$

$$= \pi(7)^2$$
$$= 153.9 \text{ cm}^2$$

as each die area is 1.55 cm^2

$$\text{Dies per wafer} = \frac{153.9}{1.55} = 99.29$$
$$\approx 99$$

② Defect per area = 0.035 cm^2

$$\begin{aligned}\text{So, Yield} &= \frac{1}{\left(1 + \left(\text{DPA} \times \frac{1.55}{2}\right)\right)^2} \\ &= \frac{1}{\left(1 + \left(0.035 \times \frac{1.55}{2}\right)\right)^2} \\ &= 0.948\end{aligned}$$

Here the yield is close to 1, so, the yield is good.

③ As we can make 10 wafers per ingot, and have 10 ingots. we make 99 die per wafer.

$$\text{So, total no of die} = 99 \times 10 \times 10 = 9900$$

~~Cost of 9900 wafers = $9900 \times$~~

$$\text{Cost per die} = \frac{13}{99 \times 0.947} = 0.138 \text{ unit}$$

$$\text{Total cost} = (9900 \times 0.138) = 1366.2 \text{ unit}$$

① As the number of die we made is equal to the order. So, we can fullfill the order.

Ans to ques - 03

① Amdahl's law means to improve a system ~~the~~ performance by doing the least tuning. For example a car might not work properly if a tire is flat. So, we just give the tire more air.

② From the given ideas, I think "making the common case faster" will help to increase the performance the most. It is something that takes least ammount of effort to improve. So, it is more related to the Amdahl's law as we are working with the common case only.

for example, while browsing on the internet there are few websites that we have to visit frequently. So, we can just bookmark them to access them faster.

Ans to ques - 04

While calculating benchmark of a system geometric mean is more preferable than the average of individual speeds because it gives us a better understanding of the system overall. Using the average method we will only get the comparison of the components, but geometric mean gives us how fast the CPU is across tested workspaces.

Ans to ques - 05

① Here, add is done = 8 times
sub is " = 5 "
mul is " = 3 "
addi is " = 2 times

total command = 18,

$$\begin{aligned}\text{total instructions} &= 8 \times 2 + 5 \times 3 + 3 \times 4 + 2 \times 5 \\ &= 16 + 15 + 12 + 10\end{aligned}$$

$$\text{Total no of instructions} = 53$$

② Here, clock cycle = 3s

$$\text{CPI} = 1$$

$$\text{no of instruction} = 53$$

$$\text{execution time} = 3 \times 1 \times 53 = 159 \text{ s}$$

$$\text{③ Clock rate} = \frac{1}{\text{Clock cycle}} = \frac{1}{3}$$

$$= 0.333 \text{ Hz}$$

④ To make the system faster we can speed up the frequent instruction. In this case it is add. If we speed up the CPI for add by 1s. the overall time reduces.

$$\text{Total instructions becomes} = 8 \times 1 + 5 \times 3 + 3 \times 4 + 2 \times 5 \\ = 45$$

As add instruction is the most frequent, I would speed up add instructions CPI.

⑤ Here, $T_{\text{unaffected}} = 5 \times 3 + 3 \times 4 + 2 \times 5$
 $= 15 + 12 + 10$

$$T_{\text{affected}} = 8 \times 2 = 16$$

As the execution time has to be reduced by 1.2 times, New execution time = $\frac{159}{1.2}$

$$= 132.5 \text{ s}$$

$$\text{NOI} = \frac{132.5}{3 \times 1} = 44.16$$

$$\text{So, } 44.16 = \frac{16}{\eta} + 37$$

$$\Rightarrow \eta = 0.4475$$

So, improvement factor for add is 0.4475.